

INDIAN INSTITUTE OF TECHNOLOGY PATNA

DBMS (MA 2205), Assignment – 3

Q1. Create a database and following table and write below queries.

Student Table

snum	sname	major	level	age
1	Alice	CS	JR	20
2	Bob	History	SR	22
3	Charlie	Math	JR	21
4	David	CS	FR	19
5	Emma	History	SO	23

Faculty Table

fid	fname	deptid
1	I. Teach	10
2	Dr. Smith	20
3	Prof. Lee	30

Enrolled Table

snum	cname
1	CS101
2	Hist1
3	Math1
1	Math1
4	CS102
5	Hist1
3	CS101

Class Table

name	meets_at	room	fid
CS101	10:00:00	R128	1
Math1	11:00:00	R130	2
Hist1	12:00:00	R128	3
CS102	10:00:00	R130	1

1. Find the names of all Juniors (Level = JR) who are enrolled in a class taught by I. Teach.
2. Find the age of the oldest student who is either a History major or is enrolled in a course taught by I. Teach.
3. Find the names of all classes that either meet in room R128 or have five or more students enrolled.
4. Find the names of all students who are enrolled in two classes that meet at the same time.
5. Find the names of faculty members for whom the combined enrollment of the courses that they teach is less than five.
6. Print the Level and the average age of students for that Level, for each Level.
7. Print the Level and the average age of students for that Level, for all Levels except JR.
8. Find the names of students who are enrolled in the maximum number of classes.
9. Find the names of students who are not enrolled in any class.